

Instructor: Date:

Learner: Regional Branch:

Program: Center:

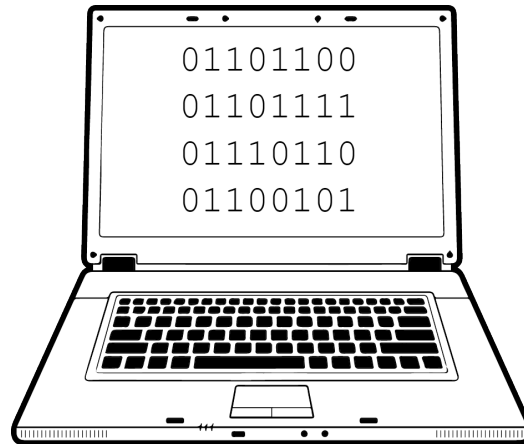
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Let's Start!

Understanding the Concepts of Values and Types



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about values and types.
- I can give detailed information about values and types.
- I can identify important information related to values and types.

Ask your partner:

- How do you think a system can find mistakes?
- What does a system do when there is a mistake?





Let's Explore!

Getting ready for an oral presentation!



Instructions: Read the text below.

Alyson: Robin, I'm finishing the last details for next week's presentation. Did you prepare your part on values and types?

Robin: Why do you ask me that? You know I am never ready! Can we have a little review before?

Alyson: Sure, I'll go first. What's a data type?

Robin: A data type is a set of values together with a set of operations.

Alyson: Correct! And don't forget what the professor told us last class about the declarative model. It always has a well-defined set of types or basic types.

Robin: Thanks for reminding me of that. I always forget important details! Now, we can also say that programs can calculate with integers or records.

Alyson: Yes, that's right! I have another question. When we have any attempt to use an operation with values, and there is a mistake, the system calculates options and raises an error condition. Why does it happen?

Robin: Because all uses of types are checked. It's impossible for a program to continue when it detects a mistake, so the program never behaves outside of the model; for instance, it hardly ever crashes due to unclear operations on the internal data structures, and it rarely raises an error condition, but it could happen.

Alyson: I'm checking here some specific information related to basic types.

Robin: Yes, I checked that yesterday and I found that the basic types of the declarative model contain numbers which have integers and floats, records which include atoms, booleans, tuples, lists, strings, and also procedures.

Alyson: But those are not all. I could also identify other elements such as chunks, functors, cells, dictionaries, arrays, ports, classes, and objects. Do you think we have to say all of this during the presentation?

Robin: Of course. And don't forget the dynamic typing.

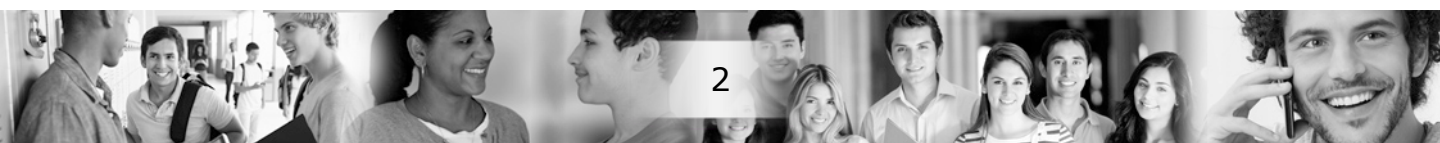
Alyson: Oh, yes! We can mention the two basic approaches to typing. I found those are dynamic and static typing.

Robin: Let me say that part. I made a chart to compare them. Well, I understood that there is a compiler which tries to verify that all operations use values of the correct type...

Alyson: That is the case of the static typing.

Robin: Right! See? You're more than ready. I told you, you need to be more confident!

Alyson: I know...





Let's Practice!

Talking About Values and Types.

Instructions: Read the sentences. Underline all the mistakes in the sentences.

1. A data type are a set of values together with a set of operations.
2. The declarative model is always important because it always have a well-defined set of types.
3. There is two basic approaches to typing: dynamic and static.



Let's Practice!

Let's check important information related to values and types.

*Instructions: Read the sentences. Write the missing word(s) in the blank. Use: **never, usually, always.***

If there is a mistake, the system calculates options and _____ raises an error condition, because a program _____ ignores a mistake when it detects it. In other words, the system _____ behaves as the instruction demands, unless it detects an error.



Extension Activity

What makes values and types important?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search further information about values and types.
- Identify aspects you consider important.
- Summarize the most relevant information in a graphic.
- Share that graphic with the class and discuss about it.

Self-Assessment

- I search for information about values and types.
- I identify important aspects.
- I summarize information in a graphic.
- I share information with my class and discuss about it.

Yes

No

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Target Words

- **behave** (verb): to act or perform in a certain way
e.g. So the program never behaves outside of the model.
- **integer** (noun): a number that is not a fraction; a whole number
e.g. Now, we can also say that programs can calculate with integers or records.
- **boolean** (noun): a binary variable that can have one of two possible values, 0 (false) or 1 (true)
e.g. Yes, I checked that yesterday and I found that the basic types of the declarative model contain numbers which have integers and floats, records which include atoms, booleans, tuples, lists, strings, and also procedures.
- **crash** (verb): to break or collapse
e.g. For instance, it hardly ever crashes due to unclear operations on the internal data structures.
- **mistake** (noun): an error
e.g. When we have any attempt to use an operation with values, and there is a mistake, the system calculates options and raises an error condition.
- **functor** (noun): type of mapping between categories that arises in category theory
e.g. I could also identify other elements such as chunks, functors, cells, dictionaries, arrays, ports, classes, and objects.



Learning Points

Simple present vs Simple Past to Contrast Events in Different Times

You use the **simple present** to talk about things that usually happen or actions that you frequently do, while **the simple past** is used to express actions that started and finished in the past.

When you express general truths and events that happen in the present we use the **simple present** tense. With this tense, we can include a time expression to specify the information, for example *every day, every week, once a month, every year, once a week*.

Don't forget that for the subject pronouns *I, we, you* and *they*, the **simple present** form is the same as the base form of the verb and for the negative and interrogative form the auxiliary verb is **do**. For the pronouns *she, he* or *it* you need to add **s** and **es**, and you need the auxiliary **does** for the negative and interrogative form.

Examples:

They design a new program.

*They **don't** design a new program.*

***Do** they design a new program?*

The company designs software every day.

*He **doesn't** design software.*

***Does** Pedro design software?*

On the other hand, you use the **simple past** tense to talk about a complete event in the past. You can form the **simple past** tense by adding *ed, -ied* to the verb when the verb is regular. For irregular verbs, however, you have to learn how they change. There are some specific time expressions you can use with this tense, for example *yesterday, last month, and two years ago*.

Examples:

We designed a new program yesterday. (regular verb)

They studied in this institution last year. (regular verb)

He found new software 5 years ago. (irregular verb)

You need the auxiliary verb **did** for the negative and interrogative form, for all the subject pronouns.

Examples:

*We **didn't** design a new program.*

***Did** they study here?*

*He **didn't** find the program.*

Don't forget to keep the verb in its base form.

Examples

- *Did you prepare your part on values and types?*
- *Why do you ask me that?*
- *And don't forget what the professor told us last class about the declarative model.*
- *When we have any attempt to use an operation with values, and there is a mistake, the system calculates options and raises an error condition.*
- *I made a chart to compare them.*



Learning Points

Adverbs of frequency, always, generally, never,

An **adverb of frequency** is a word that tells us how frequently an event happens.

Adverbs of frequency represent different percentages of regularity. The adverb never, for example, means that an action happens 0% and the adverb always expresses 100% of the times.

When you use **adverbs of frequency**, you need to consider the position of the verbs. For any verb different from **to be**, this is the structure you use:

Subject+ adverbs of frequency+ verb+ complement

For example:

We **always** send information in the morning

However, when the main verb of the sentence is the verb **to be** you follow another structure:

Subject+ verb to be +adverbs of frequency + complement

For example:

We are **always** in time for the meetings with the manager.

Take a look at the chart of **adverbs of frequency**:

Adverbs of frequency	
100%	Always
90%	Usually
80%	Generally
70%	Often
50%	Sometimes
30%	Occasionally
10%	Hardly ever
5%	Rarely
0%	Never

Examples

- *You know I am never ready!*
- *It always has a well-defined set of types or basic types.*
- *Thanks for reminding me of that, I always forget important details!*
- *For instance, it hardly ever crashes due to unclear operations on the internal data structures.*
- *It rarely raises an error condition, but it could happen.*